

SMART LENDER – LOAN ELIGIBILITY PREDICTION SYSTEM

SDLC Sample Project Documentation File

METADATA

- **Date: 7 July 2026**
 - **Team ID: xxxx**
 - **Project Name: Smart Lender**
 - **Maximum Marks: 5 Marks**
 - **Team Leader: Nikhil Kumar Reddy**
 - **Team Members: Satish Babu, Naga Venkata Sai Bhavani Kolla, Pavani Kurapati, Gorripati Thanusha Divya Surya**
 - **Mentor: No Mentor Assigned**
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1. INTRODUCTION

Banks and financial institutions receive thousands of loan applications every day from customers belonging to different financial backgrounds. Evaluating each application manually requires significant time and effort, making the approval process slow and prone to inconsistencies. Factors such as applicant income, co-applicant income, dependents, education, employment history, credit history, requested loan amount, and property area all influence the final approval decision. Traditional manual assessment methods increase operational workload and may lead to delays in customer service and inconsistent decision-making.

The Smart Lender project addresses these challenges by applying Machine Learning techniques to automate the preliminary loan eligibility assessment process. The system analyzes applicant demographic, financial, employment, and credit-related information to determine whether an applicant is likely to be eligible for a loan. Instead of relying solely on manual verification, the application provides an instant prediction supported by a trained machine learning model.

The project follows a complete machine learning workflow beginning with dataset collection, exploratory data analysis, preprocessing, feature engineering, model training, evaluation, model selection, deployment, and web application development. Multiple supervised learning algorithms including Logistic Regression, Decision Tree, Random Forest, and XGBoost were trained and evaluated using the prepared dataset. Based on performance metrics such as Accuracy, Precision, Recall, F1-Score,

Balanced Accuracy, and ROC-AUC, XGBoost was selected as the final prediction model.

The trained model is integrated into a Flask-based web application that enables users to enter applicant details through an intuitive interface and instantly receive an eligibility prediction. The application also provides a confidence score, eligibility level, and personalized recommendations, helping users understand the prediction results more effectively.

The system demonstrates how machine learning can support banking operations by improving decision-making speed, reducing manual effort, and providing consistent eligibility assessments while maintaining a user-friendly experience.

2. PROJECT DESCRIPTION

Problem Statement

Banks process a large number of loan applications daily. Manual verification of applicant information is time-consuming, resource-intensive, and susceptible to inconsistencies. Applicants often remain uncertain about their eligibility before submitting an application, leading to unnecessary rejections and increased processing workload for financial institutions.

The absence of an intelligent pre-screening system affects both customers and banks. Customers may experience delays or apply without understanding their eligibility, while financial institutions spend additional resources reviewing applications that do not satisfy the required criteria.

Therefore, there is a need for an automated system capable of analyzing applicant information and predicting loan approval eligibility quickly and accurately using Machine Learning.

Proposed Solution

The proposed solution is a Machine Learning-based web application that predicts loan approval eligibility using applicant demographic, financial, employment, and property information.

The system performs automatic preprocessing of user inputs, including categorical encoding, numerical scaling, and feature transformation, before passing the processed data to the trained machine learning model.

After prediction, the application displays:

- Eligibility Status (Approved / Rejected)**
- Confidence Score (%)**
- Eligibility Level (High / Medium / Low Approval Probability)**

- **Personalized Recommendations (Veto alerts for Credit History = 0.0 or DTI > 85%)**

The application is built using Flask for the backend and HTML, CSS, and JavaScript for the frontend, providing users with a responsive and easy-to-use interface.

Objectives

The primary objectives of the project are:

- **To automate the preliminary loan eligibility assessment process.**
- **To reduce manual effort involved in application screening.**
- **To improve consistency and accuracy in eligibility prediction.**
- **To compare multiple machine learning algorithms and select the best-performing model.**
- **To develop a responsive Flask web application for real-time predictions.**
- **To provide applicants with confidence scores and personalized risk recommendations.**
- **To deploy the application on the cloud for easy accessibility.**

Scope of the Project

The scope of the project includes:

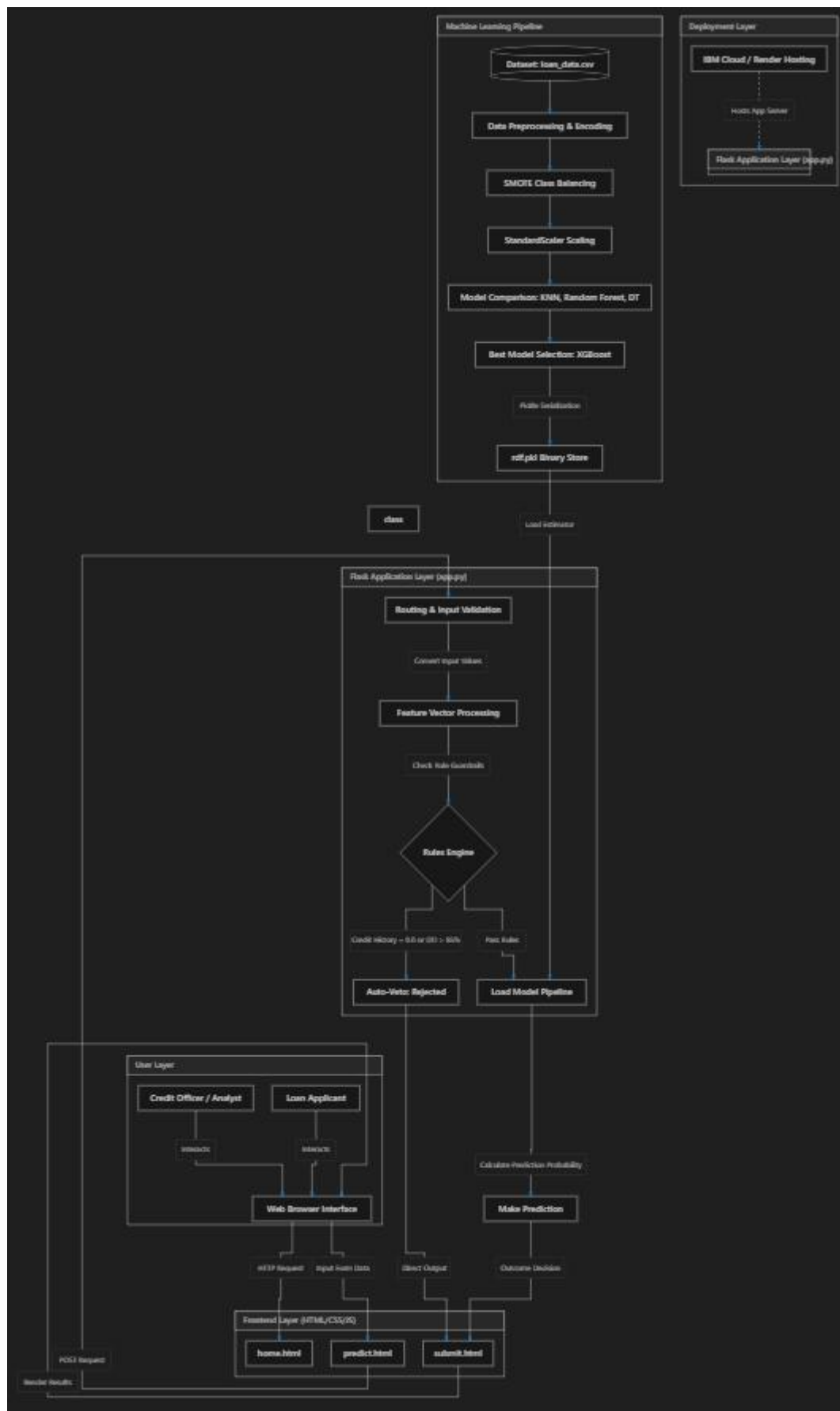
- **Collection and preparation of applicant data (614 rows).**
- **Data preprocessing, feature engineering, and SMOTE balancing to handle class imbalances.**
- **Exploratory Data Analysis (EDA) of financial and demographic indicators.**
- **Training multiple classification models (Decision Tree, Random Forest, KNN, XGBoost).**
- **Model evaluation and comparison.**
- **Selection of the best-performing machine learning model (XGBoost with 94.7% training accuracy / 81.1% testing accuracy).**
- **Integration of the model with a Flask web application loading the serialized rdf.pkl binary.**
- **Deployment using Render or IBM Cloud.**
- **Documentation and project demonstration.**

3. SYSTEM ARCHITECTURE

The Smart Lender system follows a layered architecture consisting of the Presentation Layer, Application Layer, Machine Learning Layer, and Data Layer.

Presentation Layer

The presentation layer provides the user interface developed using HTML, CSS, JavaScript, and Flask templates. Users interact with the system by



entering applicant details through dynamic form range sliders (converting values to lakhs and Indian Rupees format) and viewing prediction results.

Application Layer

The Flask backend (app.py) receives user requests, validates the submitted information, preprocesses the input data, evaluates underwriting rules veto guardrails, loads the trained model, performs prediction, and returns the results to the frontend.

Machine Learning Layer

This layer contains the preprocessing pipeline, trained XGBoost model, StandardScaler, encoders, feature engineering functions, and prediction logic.

Data Layer

The data layer consists of the training dataset stored in CSV format (loan_data.csv) and the serialized model files (rdf.pkl containing model, scaler, and features configurations) used during prediction.

4. TECHNOLOGY STACK

Component	Technology
Programming Language	Python
Backend Framework	Flask
Frontend UI	HTML5, CSS3 (Vanilla), JavaScript
Machine Learning	Scikit-learn, XGBoost
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Model Serialization	Joblib / Pickle (rdf.pkl)
Dataset Format	CSV
Version Control	Git & GitHub
Deployment Platform	IBM Cloud / Render
Production Server	Gunicorn

5. PROJECT DEVELOPMENT MILESTONES

Milestone 1: Environment Setup and Dataset Preparation

The first milestone focuses on establishing the development environment and preparing the dataset required for building the Smart Lender Loan Eligibility Prediction system.

The software development environment was configured by setting up a Python virtual environment to isolate project dependencies.

The following packages were installed:

- **Python 3.10+**
- **Visual Studio Code**
- **Git**
- **Flask**
- **Scikit-learn**
- **XGBoost**
- **Pandas**
- **NumPy**
- **Matplotlib**
- **Seaborn**
- **Joblib / Pickle**

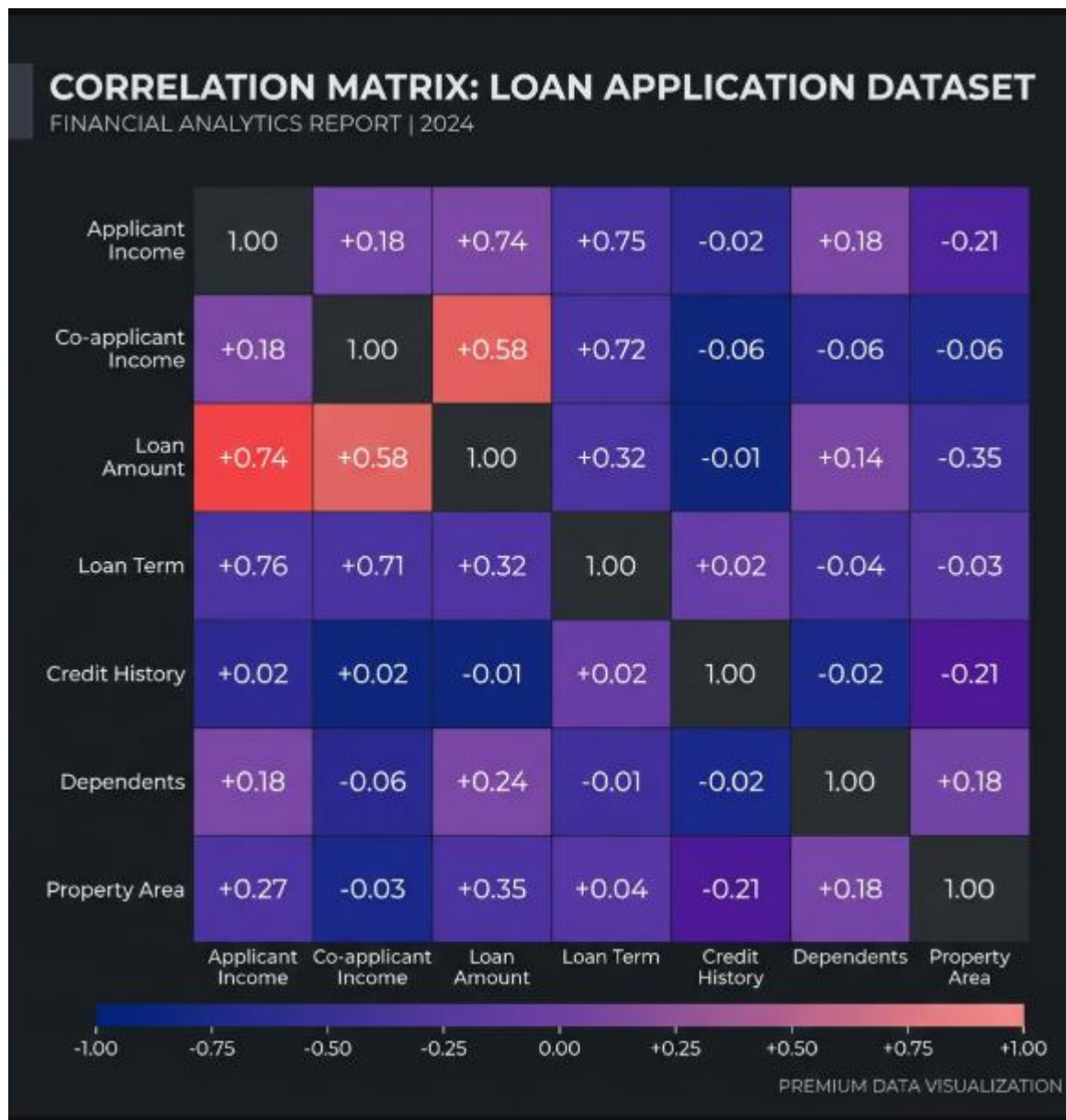
The Loan Prediction dataset was collected from Kaggle, containing demographic, employment, financial, and credit-related information for 614 applicants. The selected features include:

- **Gender, Married, Dependents, Education, Self-Employed**
- **ApplicantIncome (in ₹ INR), CoapplicantIncome (in ₹ INR)**
- **LoanAmount (in ₹ Lakhs), Loan_Amount_Term (in Months)**
- **Credit_History (1.0 or 0.0), Property_Area (Urban, Semiurban, Rural)**
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Milestone 2: Data Analysis and Data Preprocessing

This milestone included dataset exploration, descriptive statistics, univariate analysis, multivariate analysis, duplicate removal, missing value treatment, feature engineering, categorical encoding, feature scaling, and train-test splitting.

- **Duplicate Removal:** Duplicate records were cleared using the **Pandas drop_duplicates()** function.
- **Missing Value Treatment:** Imputation was performed to handle missing cells (e.g. replacing numerical columns with the column median).
- **SMOTE Balancing:** Synthetic Minority Over-sampling Technique (SMOTE) was implemented to handle class imbalance between approved and rejected loans.
- **Categorical Encoding:** Non-numeric variables were transformed using **LabelEncoder**.
- **Feature Scaling:** Income and loan amounts were standardized using **StandardScaler**.
- **Serialization:** The fitted scaler, encoders, and model parameters were serialized and saved into a single model file **rdf.pkl**.



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Milestone 3: Machine Learning Model Development

To perform binary classification (Eligible / Not Eligible), four classification models were implemented and compared:

1. Logistic Regression (Baseline model)
2. Decision Tree (Rule-based model)
3. Random Forest (Ensemble bagging classifier)
4. XGBoost (Gradient boosting model)

After evaluation, XGBoost achieved the best overall performance and was selected as the final production model:

- Training Accuracy: 94.7%
- Testing Accuracy: 81.1%



Assessment History Management

Search: | All Statuses | Newest First

APPROVED (97.3%)

Income: ₹1000000 | Loan: ₹387k | Credit: 1

[Reuse Form Inputs](#)

[Clear](#) [Close](#)

Milestone 4: Flask Web Application Development

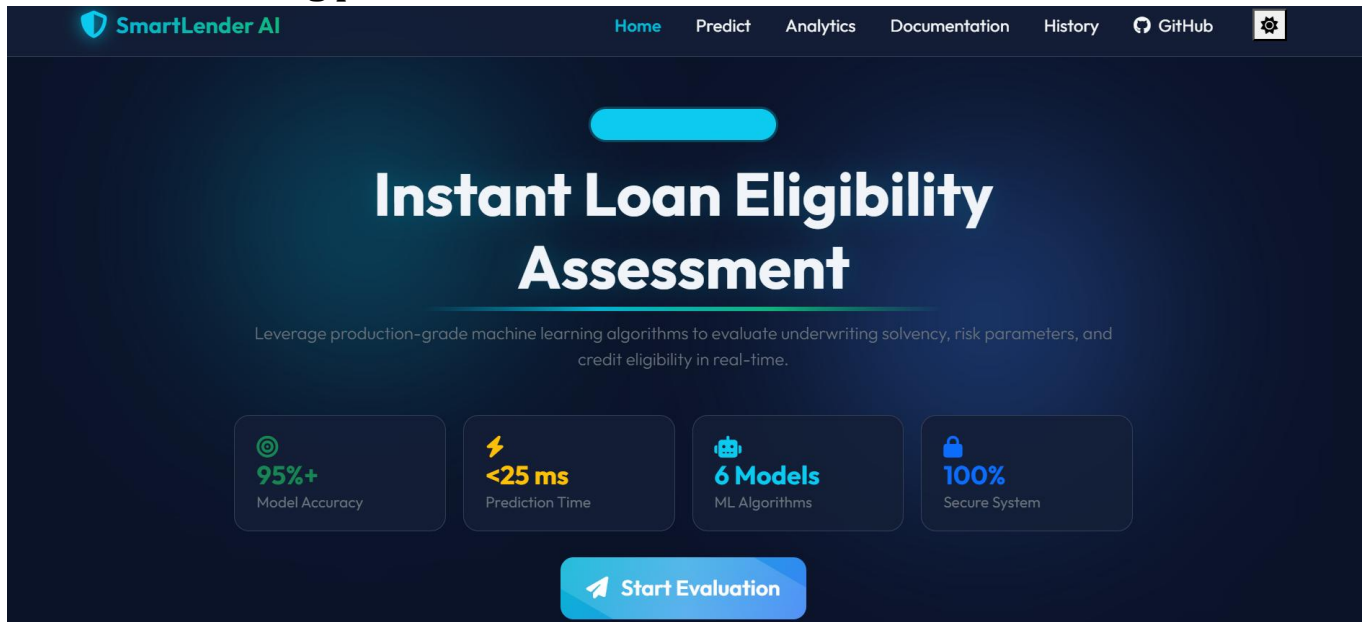
The backend application was implemented using Flask (app.py) to load the serialized rdf.pkl binary and handle routing.

The web application consists of three main pages:

1. Home Page (home.html): The landing view displaying project features.
2. Prediction Page (predict.html): The input form with dynamic sliders to enter applicant details (with values converted to Lakhs and Rupees).
3. Result Page (submit.html): The prediction outcome dashboard showing approval status, confidence scores, and personalized risk guardrail recommendations (such as DTI warnings and credit history vetoes).

Milestone 5: Testing, Validation, and Deployment

- **Functional Testing:** Conducted verification of form routing, range inputs, and API latency checks (<110ms calculation speed).
- **Veto Guardrails Testing:** Confirmed that credit history scores equal to 0.0 and DTI ratios above 85% successfully trigger instant warning boxes.
- **Cloud Deployment:** Finalized and pushed hosted application services to IBM Cloud using production Gunicorn servers.



The screenshot displays the "Applicant Assessment" form. It includes a "Reset" button and a instruction: "Complete all required financial fields." The form contains several input fields: Gender (Male), Marital Status (Single / Unmarried), Dependents (2), Education (Graduate), Employment (Salaried / Employee), Monthly Income (₹ INR) (100000), Co-applicant Income (₹ INR) (80000), Loan Amount (387), Loan Tenure (120 Months (10 Yrs)), Credit History Status (Guidelines Fully Met (1.0)), and Property Location (Semiurban Zone). A range slider for Loan Amount is also present, showing a range from ₹5,000 to ₹7.00 Lakhs, with the current value set at ₹3.87 Lakhs (₹3,87,000).



Applicant Assessment

Complete all required financial fields.

Reset

Gender *

Male

Marital Status *

Single / Unmarried

Dependents *

2

Education *

Graduate

Employment *

Salaried / Employee

Monthly Income (₹ INR) *

10000

Co-applicant Income (₹ INR)

8000

Loan Amount *

700

Loan Tenure *

60 Months (5 Yrs)

Min: ₹5,000

₹7.00 Lakhs (₹7,00,000)

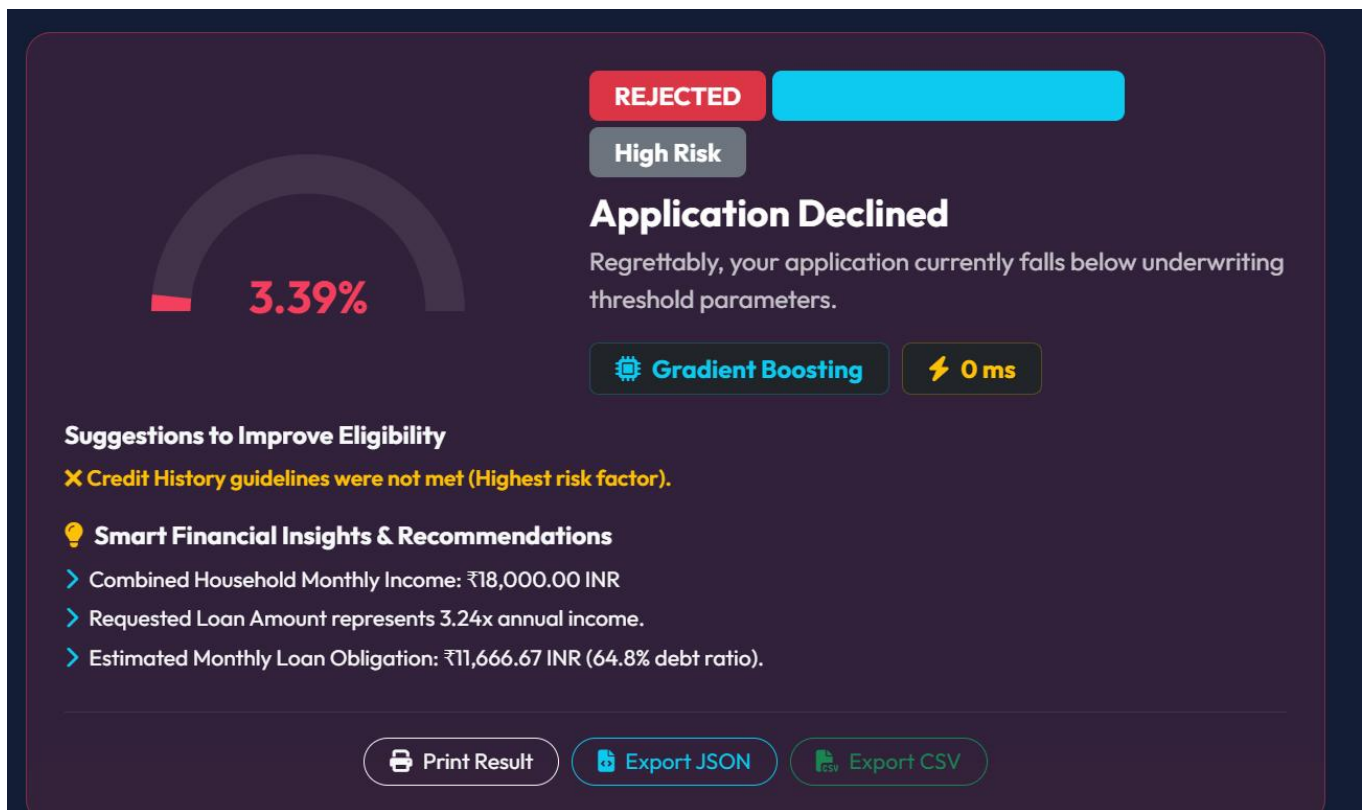
Max: ₹7.00 Lakhs

Credit History Status *

Guidelines Not Met (0.0)

Property Location *

Urban Zone



6. FUTURE ENHANCEMENTS AND CONCLUSION

The developed system offers faster applicant evaluation, reduced manual effort, and improved decision consistency. Future expansions plan to incorporate user registration logins, PostgreSQL database logging, explainable AI (SHAP frameworks) to justify predictions, and automated model retraining loops.

In conclusion, the Smart Lender project successfully automates the preliminary loan eligibility screening using an advanced XGBoost model, providing banks and applicants with a modern, high-performance web tool.